

Interplanetary CubeSat missions

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*Benjamin K. Malphrus^a, Anthony Freeman^b, Robert Staehle^b,
Andrew T. Klesh^b, and Roger Walker^c*

^aMorehead State University, Space Science Center, Morehead, KY, United States,

^bJet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States,

^cEuropean Space Agency, Noordwijk, The Netherlands

1 Introduction

A new era of Solar System exploration began on May 5, 2018 with the launch of the twin Mars Cube One (MarCO) CubeSats. The success of the MarCO mission, along with 13 interplanetary CubeSats planned for launch on Artemis 1 in 2021, is establishing a new paradigm of planetary exploration. This new paradigm includes CubeSats and other small satellite platforms in supporting roles like MarCO and in primary roles like the Artemis 1 Lunar Cubes (Lunar IceCube, Lunar Flashlight, and LunaH-Map). Numerous studies are underway in the United States and Europe, in particular, that utilize CubeSats and CubeSat constellations that have the potential to revolutionize robotic Solar System exploration. This chapter examines the missions underway and the upcoming missions and discusses the differences between low Earth-orbiting CubeSats and interplanetary CubeSats. The chapter ends with a projection of future Solar System exploration made possible by this new standard.

Interplanetary CubeSats take advantage of the CubeSat paradigm and of the availability of commercial components developed for low Earth orbit (LEO) missions, but they are specifically designed to explore deep space. As a result, interplanetary CubeSats are essentially very different from low Earth orbit CubeSats in three primary technological areas: propulsion, radiation tolerance, and telecommunications. While fundamental differences exist in these areas, interplanetary CubeSats require changes to almost every satellite subsystem. Interplanetary CubeSats require propulsion systems capable of generating enough ΔV to support trajectories from Earth escape or from GEO transfer to interplanetary destinations. Interplanetary CubeSats often need power systems with lower power modes and higher energy storage capabilities since they have a higher level of power requirements than LEO missions, due to the presence of propulsion and due to demanding telecommunication systems. Due to higher power level requirements, more advanced thermal control solutions are often needed to dissipate excess heat to deep space. Interplanetary CubeSats also require strategies for radiation tolerance as they are outside the protection of Earth's magnetosphere. Attitude determination and control systems (ADCS) for interplanetary CubeSats need a combination of traditional control system and propulsion to avoid the issues of wheel saturation outside the Earth's geomagnetic field. In terms of autonomy, interplanetary missions have

less frequent contact with the ground than LEO missions, imposing the need of agile algorithms to facilitate autonomous onboard operations. Finally, one of the most important differences between LEO missions and interplanetary missions is represented by the telecommunication systems. Telecommunication systems for interplanetary CubeSats face harsher environments and longer path distances and have more sophisticated navigation needs than the LEO CubeSats. For this reason the design of telecommunication systems for interplanetary missions is extremely challenging, and significant development is currently ongoing in the areas of radio design, antenna design, and in the design of ground support architectures. Optical communications have also been considered for interplanetary CubeSats, though that technology is less mature. In general, interplanetary CubeSats require different and more complex spacecraft systems architectures and must utilize different and more sophisticated ground station systems than LEO missions [1]. These differences are discussed in this chapter in detail using specific examples from current and planned missions.

Nine years after the idea was first floated in a proposal to NASA's Innovative Advanced Concepts (NIAC) opportunity [2], we are on the threshold of a new era in robotic exploration of our Solar System, one in which CubeSats and SmallSats will play an important role [3,4]. As flight opportunities begin to open up, deep space CubeSats and SmallSats could exhibit exponential growth, as seen for CubeSats in low Earth orbit [5]. Indeed, growing interest has been expressed within the space and Earth science community for such missions [6]. For Earth-orbiting CubeSats and SmallSats, the growth trend is driven by the push toward constellations, which may consist of multiple elements that each yield unique measurements that collectively serve one higher objective [7], or of similar elements that offer significantly enhanced temporal resolution [8,9], consistent with recently expressed requirements from the Earth science community [10]. It is also clear that miniaturization of many key spacecraft technologies across a broad front will have a profound effect on future deep space exploration [11], projecting out into the far future, perhaps even on interstellar exploration [12]. It is these kinds of projections that stimulate interest within the space science community for interplanetary CubeSat and SmallSat missions.

2 Destinations

If CubeSats are to be sent out into unexplored corners of our Solar System, where can they reasonably be expected to roam? Fig. 1 illustrates the regions that are most easily accessible and for which SmallSat missions are operating or planned. The inner Solar System is considered reachable with both free-flying and mother-daughter configuration CubeSats/Nanosats, while the outer Solar System is currently compatible with just the mother-daughter or ride-along configuration. Without a suitably compact radioisotope power source, deep space nanosats heading out to Jupiter and beyond will likely have to hitch a ride along with their host spacecraft, run on energy stored in batteries after their release from their storage/hibernation container, and relay data back to Earth through the mother spacecraft. Beyond the CubeSat form factor,

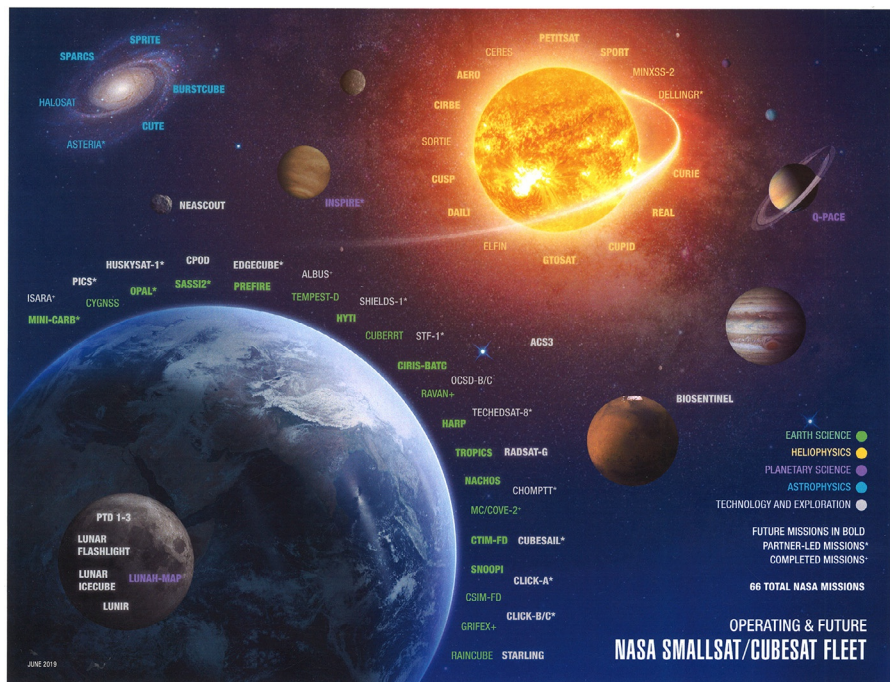


Fig. 1 Exploring our Solar System with CubeSats and nanosats.
Image Courtesy of NASA/JPL/Caltech.

but within the capability of a SmallSat on an ESPA ring, a NASA Innovative Advanced Concepts (NIAC) investigation was initiated utilizing an inflatable solar concentrator/telecomm antenna up to 5 m in diameter, low duty cycle operations, very low-power system-on-a-chip (SoC) electronics, and other advancements to enable heliophysics or flyby missions out to the orbit of Neptune or beyond [12a]. These spacecraft might be carried as secondary payloads on launches of future primary missions to Jupiter or Saturn.

In recent decades, several deep space missions have used a gravity assist or slingshot maneuver at Venus to provide a boost in velocity to propel them on their way to their primary mission target. These missions include Galileo (Jupiter), MESSENGER (Mercury), Cassini (Saturn), and Solar Probe (the Sun). Venus gravity assists have become an arrow in the mission designer's quiver that is pulled out whenever we need to save time, money, or propellant to get where we want our primary mission to go. Now imagine that every time a mission executes a gravity assist at Venus, it could drop off a small nanosat probe that could sample and analyze part of Venus' atmosphere, relaying the data back to Earth via the vehicle that dropped it off. Straight shot, independent trajectories to Venus are also possible for nanosats and could launch as secondaries on a launch vehicle propelling a larger spacecraft on a Venus trajectory or with a hybrid approach involving a number of nanosats being carried on a larger microsat carrier platform that is itself released by the launch vehicle. With its thick

atmosphere, Venus is a good target to demonstrate aerocapture with a nanosat, placing into orbit around our sister planet.

Closer to home, the Moon is an attractive destination for nanosats, since it can be reached in a relatively short time, and communication distances are manageable. Near-Earth asteroids (between 0.98 and 1.3 AU distant from the Sun) represent another rich target set, with their number estimated at over 14,000, and ~ 1000 of them larger than 1 km in diameter. Larger mothercraft carrying several daughter CubeSats could be used for deep in situ multipoint characterization of individual asteroids. Due to cost, however, it is unlikely that with this approach, we will ever visit more than a very few with a NASA Discovery-class mission like NEAR Shoemaker [13] or the Japanese Hayabusa missions [14]. But if these asteroids could be reached with CubeSats, it may become possible to examine hundreds of them close up within a generation, at a price tag comparable with that of a single Discovery mission. As a first step in this direction, the MSFC-JPL Near-Earth Asteroid Scout (NEAScout) CubeSat mission is slated for launch in 2021 aboard the same Artemis launch that will carry the first lunar CubeSats.

Launch periods for energy-efficient Mars transfers occur at intervals of about 26 months, and in recent times, most of these launch periods have had one or more spacecraft inserted onto a trajectory toward the red planet. Nanosats could ride along as ballast to make up the launch mass on larger Mars missions or make their way there as free-flying spacecraft, as the twin MarCO spacecraft did [15]. With a few kilometers per second of ΔV capability, free-flying nanosats could be placed into orbit around Mars, opening up the potential for science from orbit at dramatically lower cost and adding to the communications and navigation infrastructure.

The main asteroid belt, situated between 2.2 and 3.2 AU (AU) away from the Sun, contains between 0.7 and 1.7 million objects with a diameter > 1 km. Thus far, spacecraft en route to the outer planets have given us a closer glimpse of only a handful, and the Dawn mission [16] has provided detailed examination of just two of the largest: Ceres and Vesta. This is definitely rich hunting ground for CubeSat missions—the asteroid belt is close enough to the Sun that solar power generation is practical and close enough to Earth that communication distances are not extreme. It can take a long time to get out to the main asteroid belt though—Dawn arrived at its first target, Vesta, nearly 4 years after launch—so reliability and longevity will need to be demonstrated over several years if nanosat missions to the main belt are to succeed.

There are also more than 5000 known comets, whose orbits have taken them close enough to the Sun to be visible from Earth. Again, our spacecraft have visited less than a dozen: Halley's comet, 21P/Giacobini-Zinner, 26P/Grigg-Skjellerup, 107P/Wilson-Harrington, 19P/Borrelly, 81P/Wild, 9P/Tempel, 103P/Hartley, 2P/Encke, and 67P/Churyumov-Gerasimenko, which leaves numerous possibilities for nanosat mission flybys and encounters of future comets. Enough is known about comet trajectories to plan and quickly execute flyby missions, but rendezvous missions take longer and will require a higher standard of reliability: ESA's flagship Rosetta mission took 10 years to catch up with comet 67P/Churyumov-Gerasimenko and match trajectories [17]. Such long mission durations are not unusual when missions are planned to chase the so-called "short-period" comets.

Or a strategy could be implemented to allow the outer Solar System and beyond come to us. CubeSat missions are uniquely positioned to take advantage of the expanding reach of ground-based telescopes that can now detect objects such as Oumuamua [18] and the Oort cloud comet CK 2017 [19] several years before they traverse the inner Solar System. Given sufficient advance warning, fast response missions that offer a close encounter of these objects become feasible. CubeSats can have much shorter development cycles than conventional spacecraft (MarCO was put together and made ready for flight in under 18 months); all we need is to marry that capability with one of the new low-cost launch vehicles, capable of achieving a high C3. These strategies will further accelerate the interplanetary CubeSat paradigm.

3 What makes interplanetary CubeSats different?

Succinctly put, interplanetary CubeSats are different because their orbits take them far from home, our mother planet Earth, and its familiar environment. To be successful in low Earth orbit, a CubeSat must be relatively complex in a manner that affords powering and survival of electronics and other equipment, maintaining specified orientations, and responding to commands, all while acquiring and transmitting health status and data to meet mission objectives. Once a CubeSat's orbit leaves Earth's vicinity, all these functions must be provided, but for some of these functions, *how* to do so requires new techniques and complexity. For most CubeSats in Earth orbit, and especially LEO, getting to the mission's destination orbit is fundamentally over within minutes when the CubeSat starts operating. For most interplanetary CubeSats, most of the mission and its complexity will be exercised while transiting from the mission's launch orbit to one or more destination orbits where the CubeSat can perform its intended mission by acquiring data unavailable from elsewhere.

The following topics will all be covered later in this chapter because they represent significant differences from LEO CubeSat missions: (a) operating environment dominated by the need for radiation tolerance, compounded by mission duration; (b) telecommunications, which is driven especially by the distances involved; (c) navigation to a variety of destinations; (d) instrumentation, some of which differs from that common in LEO; (e) onboard data storage and processing often required because telecommunication distance necessitates lower data rates and longer latencies; (f) operations autonomy driven by lack of frequent commanding opportunities and lack of frequent visibility into onboard housekeeping parameters; and (g) transit to a destination as a secondary payload.

Recognizing that every mission is unique, some other aspects of interplanetary spacecraft and missions remain much the same as for any orbit, including (a) structure and mechanical systems, (b) thermal control, (c) some forms of attitude determination and control, and (d) propulsion. With respect to propulsion, some beyond-Earth CubeSat missions have extremely demanding total ΔV requirements that put them in a class outside of what may be required for any mission in Earth orbit.

4 Historical perspective and the first interplanetary CubeSat developers

By the start of 2011, over 50 CubeSats had been launched, all to low Earth orbit, with 3U being the largest form factor. By 2019, over 1000 CubeSats had been launched [20]. In the early days of the CubeSat initiative, very few rigorous scientific investigations were performed, and the form factor was not generally regarded as being highly useful except as a university tool, and there was no serious discussion of going beyond Earth orbit. A few people suspected that this situation would change. For those who even thought about it, surely there were obstacles. For one thing, no credible telecommunication package existed with the required small size (e.g., a 10-cm cube or one-third of the then-largest CubeSat), mass (~ 1 kg), and power (< 10 – 20 W) that could send data or receive commands beyond a few thousand kilometers. For another, the published “CubeSat Standard” prohibited “propulsion” from being aboard CubeSats carried as secondary payloads, because to most people, this term implied the use of stored and potentially unstable chemical energy that without extensive qualification could represent a threat to the primary payload and its launch if something went wrong inside the ride-along CubeSat. So, even if a way to communicate were established, how would the small spacecraft get anywhere interesting? Further, many of the electronic components readily used in low Earth orbit were thought to be intolerant to radiation levels commonly found beyond the shielding cocoon of Earth’s magnetosphere. Radiation-tolerant electronics were well known, but often beyond the budget of most CubeSat implementors.

After 2010, some factors began to converge that inevitably would have led to interplanetary CubeSats at some point. But it would be difficult to overemphasize how preposterous the concept seemed to most space cognoscenti at the time. The most basic enabling factor was that CubeSat size was on a path toward the 6U form factor. Two propulsion techniques were being developed and miniaturized that had high ΔV capability without any unusual stored energy during launch: solar sails and electric propulsion not requiring pressurized propellants. At JPL, both optical and RF telecommunication experts were working to dramatically reduce the size of needed electronics.

Robert Staehle at JPL was inspired by the work of longtime colleague Tomas Svitek of Stellar Exploration, Inc. on the solar sail for The Planetary Society’s LightSail[tm]-1 and asked how big a sail might be deployable from 2U of a 6U CubeSat. It turned out that such a sail, ~ 9 m on a side, would be able to impart > 1 km/s of ΔV per year of mission duration at 1 AU. Consulting with telecom experts Hammid Hemati, then with JPL, and Courtney Duncan, also at JPL, indicated that they believed they were on technology development paths toward 1U equipment that could provide > 1 kbit/s from Mars using optical and RF links, respectively. Pantazis Mouroulis in JPL’s Instruments Division showed that a new optical layout could reduce the shoebox-sized Moon Mineralogy Mapper that had been provided to India’s Chandrayaan-1 lunar orbiter to 2U with similar performance as an imaging spectrometer covering the visible-to-IR wavelength range 400–3000 nm. Armed with this information at the end of 2010, plus a few back-of-the-envelope calculations, Robert Staehle met with

CubeSat coinventor Jordi Puig-Suari of California Polytechnic University, San Luis Obispo, to show that with 2U for an instrument payload, 2U for propulsion, 1U for telecommunications, and 1U for everything else, meaningful interplanetary CubeSat missions would be possible. Jordi and other team members were assembled, and their 2011 proposal to NASA's just-reestablished Innovative Advanced Concepts (NIAC) was submitted, selected, and funded to define a convergent technology path to enable an example CubeSat mission to visit asteroids riding as a secondary payload to be deployed off a launch to GEO or escape.

At the same time, the NIAC investigation group asserted the feasibility of a wider set of example missions serving different disciplines [21]: (1) Mineral Mapping of Asteroids (*Small Body Science*); (2) Solar System Escape Technology Demonstration (*Tech Demo*); (3) Earth-Sun Sunward of L1 Space Weather Monitor (*Heliophysics and Terrestrial Applications*); (4) Phobos Sample Return (*Planetary Science*) (Fig. 2); (5) Earth-Moon L2 Radio Quiet Observatory (*Astrophysics*); and (6) Out-of-Ecliptic Missions, for example, Solar Polar Imager CubeSat Constellation (*Heliophysics*); (7) Lunar Surface Water Ice Mapper (Lunar Science & Resource Exploration); and (8) Near-Earth Asteroid NanoSat Lander.

Before their study report was complete, the group proposed a more detailed analysis of two particularly difficult missions that just might be possible with CubeSats: a sample return from Phobos using *two* CubeSats suggested by team member Louis Friedman, and observing the Sun's poles from a heliocentric latitude of 45 degrees, attained using an interplanetary superhighway trajectory conceived and later mapped out by coinvestigator Martin W. Lo. Many observers still consider these concepts preposterous.

At the AIAA Space 2012 conference in Pasadena, California, the NIAC group recommended that "NASA could enable dramatic new capability by making launch

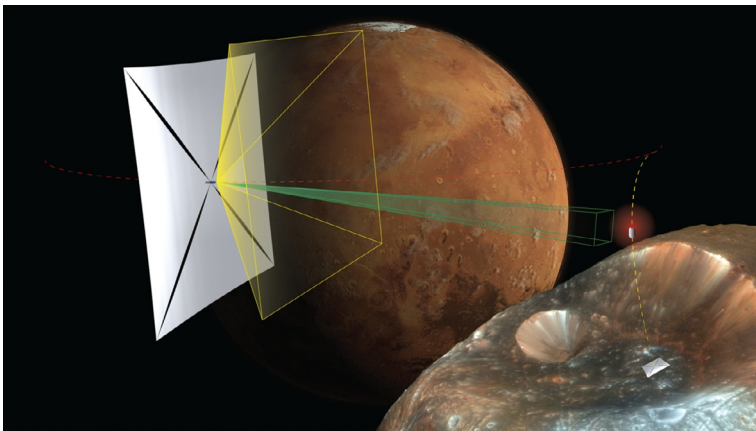


Fig. 2 Far-out concept: Interplanetary CubeSats retrieving a sample from Phobos or Deimos. Artist's concept by Ryan Sellers/CalPoly-SLO, courtesy of NASA/JPL/Caltech.

slots and funding available to support CubeSats on all launches to C3 $>\sim 0$, and as hosted riders aboard some fraction of geostationary satellites” [22]. NASA later adopted a similar policy at the heart of their Small Innovative Missions for Planetary Exploration (SIMPLEx) program [23].

In response to lunar Strategic Knowledge Gap goals announced by NASA HEOMD, Robert Staehle, David Eisenman, and Garry Burdick kicked off teams at JPL to propose an asteroid-visiting CubeSat and the Lunar Flashlight mission concept (with MSFC) now slated to fly aboard EM1, recently redesignated as Artemis 1. At the same time, Les Johnson of MSFC formed a team to independently propose a solar sail-propelled multiasteroid mission remarkably similar to the JPL proposal, which HEOMD combined upon selection to create the NEA Scout mission, also slated for EM1. Simultaneously, Ben Malphrus (Morehead State University) formed a team including GSFC to propose Lunar IceCube, also now slated for Artemis 1. A short time after that, Craig Hardgrove of Arizona State University proposed LunaH-Map into NASA’s first SIMPLEx call for proposals also now slated for Artemis 1.

As evidenced by this multiplicity of proposals with broad participation, funded and in various stages of development, the *potential* value of interplanetary CubeSats had been asserted convincingly to enough people with enough influence to secure the necessary resources. It remained only to clearly *prove* the value in flight to complete the transition from nearly total ignorance and skepticism to acceptance, if not full embrace, by the community.

Leadership of that step fell to Andrew Klesh, Joel Krajewski, and the team John Baker helped them to assemble for the Mars CubeSat One (MarCO) mission, which gained quick support from JPL’s director, Charles Elachi, and the Assistant Director at that time, Firouz Naderi. Started in late 2014 when the twin Interplanetary NanoSpacecraft Pathfinder In a Relevant Environment (INSPIRE) mission had been passed up for a near-escape launch opportunity, MarCO was on a fast schedule to meet the launch date for the Discovery InSight mission in early 2016, with which the two MarCOs were to ride on the same launch to Mars. In spite of the rapid schedule, the team secured the resources to leapfrog INSPIRE’s capabilities with new electronics and other innovations to make the design capable of all the fundamental functions of planetary flight found in missions costing hundreds of millions of dollars, all for well less than \$20M [24].

MarCO had to be delayed because the launch of the primary mission to carry it—Interior Exploration using Seismic Investigations, Geodesy and Heat Transport (InSight)—had to be delayed 26 months because *its* payload was not ready on time. Once launched, MarCO succeeded in every way planned at the outset, with the A and B spacecraft both performing their real-time “bent-pipe” relay of communications throughout InSight’s landing and even taking pictures of Mars. Thus in the span from 2010/11 to 2018, MarCO’s success (see [Section 6.1](#)) swept away much of the prior skepticism and outright opposition to using CubeSats and, by extension, SmallSats, for planetary science and exploration. The door to CubeSat exploration of the Solar System was opened.

5 Solution paths to uniquely interplanetary challenges

5.1 Radiation tolerance and mission duration

Higher single-event upset rates were expected on the first interplanetary CubeSats, dealt with using latchup immune parts (having linear energy transfer [LET] $> \sim 35\text{--}70\text{ MeV/mg-cm}^2$), robust software, and the ability to implement periodic power cycles with the system without disrupting normal operations. In practice, single-event upset rates have proven to be much less than predicted, perhaps because of careful part selection.

For commercial electronics, part-to-part radiation performance variability is a major challenge, because the same parts from different fabrication runs can have very different radiation performance characteristics. Ensuring mission success can imply the need to purchase enough parts from a single manufacturing run to perform radiation testing on some samples and being prepared to reject all parts from that lot in the event of test failure.

Total ionizing dose (TID) tolerance levels are lower for commercial electronics, but simple shielding techniques are sufficient since interplanetary radiation environments are generally more benign than those in GEO. There are, however, some interplanetary environments that experience intense radiation, such as near the orbits of the inner Galilean satellites of Jupiter, where radiation levels will necessitate both heavy shielding and radiation hard parts that are outside the typical CubeSat experience base.

Additional robustness is achievable using asymmetric connections among different CubeSat functions and processors; radiation-tolerant features including triple-mode redundancy, monitoring, and rebooting systems utilizing watchdog timers; the use of FPGA-based processor nodes; spot shielding; and numerous other techniques. New technologies are under development including atomic number (Z) Grade technology that shows promise for radiation shielding that will improve the performance of interplanetary CubeSats.

5.2 Propulsion systems for interplanetary CubeSats

The spectacular success of NASA/JPL's MarCO—the first interplanetary CubeSats—has raised expectations for SmallSat missions that can explore our Solar System at much lower cost than a conventional NASA mission. But MarCO only flew to Mars: the 40 m/s ΔV provided by its cold gas propulsion system fell far short of the ΔV needed to get into the orbit around the red planet (Table 1). The planetary science community's appetite for SmallSat missions will be further whetted by the upcoming Artemis 1 mission, the first launch of the SLS rocket, planned for 2021, which will send a total of 13 CubeSats on a trajectory toward the Moon. Two of them, Lunar Ice-Cube and LunaH-Map, will carry a Busek RF ion propulsion system, imparting enough ΔV ($\sim 0.8\text{ km/s}$) to place these 6U CubeSats into highly elliptical lunar orbits.

Table 1 Nominal ΔV in kilometers per second required for some commonly considered deep space CubeSat missions.

Nominal ΔV values for CubeSat deep space missions (km/s)	From LEO at 250 km	From GTO	From Earth escape
Earth Escape, for example, for Earth-trailing orbit, or L1/L2	3.2	0.7	–
Lunar orbiter at $h=100$ km	4.0	1.5	0.8
Deimos orbiter at $h=1$ km	5.3	2.8	2.1
Mars orbiter at $h=200$ km	5.8	3.3	2.6
Space weather mission at Earth-Sun L5	5.8	3.3	2.6
NEO asteroid rendezvous at 1.2 AU	6.3	3.8	3.1
Mercury flyby	5.6	3.1	2.4
Venus orbiter at $h=400$ km	6.9	4.4	3.7
Main belt asteroid rendezvous at 2 AU	9.5	7.0	6.3

Note that these values are for ballistic trajectories; values for low-thrust missions may be higher.

How far out into the Solar System is it reasonable to expect an independently flying CubeSat to venture? Table 1 provides some ΔV values for several examples. The commercially available 1 kg/1U IFM nanotruster from Enpulsion can provide enough total impulse for a 6U CubeSat to achieve 0.8 km/s. The Busek BIT-3 can propel a 14-kg 6U CubeSat to achieve 1.2 km/s of ΔV [25]. Clustered into a package of six identical units, this could be increased to stretch to a range of 2.5–3.4 km/s, again for a 6U CubeSat, although for this example the propulsion units would add another 6U to the spacecraft volume, scaling it up to a total of 12U. This range of ΔV values puts most of the inner Solar System within reach from an Earth escape trajectory and a few targets within reach from an initial geostationary transfer orbit (GTO-highly elliptical Earth orbit). From an initial LEO orbit at 250 km altitude, it may just about be possible to achieve Earth escape, enabling missions that we may want to go to an Earth-trailing orbit, or to one of the Earth-Moon or Earth-Sun Lagrangian points.

For some segments of some missions, significantly lower ΔV s can be effective, at the expense of longer transit times, by utilizing the so-called interplanetary superhighway of multibody gravitational effects, as modeled using LTool software [26–28].

5.3 Overcoming telecommunication challenges

Telecommunication issues for interplanetary CubeSats are extreme. Interplanetary CubeSats face harsher environments, have much longer path distances, and have more navigation needs than LEO CubeSats. For this reason the design of telecommunication systems for interplanetary missions is extremely challenging, and significant development is currently ongoing in the areas of radio design, antenna design, and

the design of ground support architectures. All three of these aspects of a space communication system are being addressed, and the success of the MarCO mission shows that the telecommunication challenges faced by interplanetary CubeSats can be overcome with available technologies and proper mission design.

UHF frequencies that are typical of LEO CubeSats can be used, to some extent, at lunar distances, but getting a signal back to Earth at a reasonable data rate, and sending commands up to the spacecraft, generally requires higher frequencies to limit free space losses. While UHF systems may still have value for proximity operations of interplanetary CubeSat constellations, higher frequencies (S-band, X-band, and Ka-band) are more appropriate. To close a telecommunication link over interplanetary distances requires higher Equivalent Isotropic Radiated Power (EIRP) on the spacecraft and the use of large apertures with sensitive electronics on the ground. Since output power of deep space transceivers is limited (typically 4–7 W), the only degrees of freedom related to the communication design are wavelength and antenna dimensions (although effectively higher throughput can be achieved using higher-order modulation schemes, encoding, and advanced techniques like delay tolerant networking). Regarding antennas, typically slotted waveguide, patch or parabolic antennas are used at the higher frequencies, since gain is a function of size. Examples of high-gain antennas flown on CubeSats include the reflectarray antennas on MarCO (0.3×0.6 m X-band antenna) and the ISARA 0.34×0.24 m Ka-band reflectarray. NASA's RainCube flew a successful deployable 0.5-m-diameter Ka-band parabolic antenna. These missions have proven that the relatively high-gain antennas needed for deep space missions can be adapted to the CubeSat platform [29].

Data rates achieved by large interplanetary spacecraft on deep space missions are a function of distance given that they typically use the most effective technologies (high-gain antennas and the use of the NASA Deep Space Network). NASA's Mars Reconnaissance Orbiter (MRO) achieves a data rate of 4 Mbps from Mars. Juno at Jupiter's distance is reduced to 200 Kbps, while Rosetta at comet 67P/Churyumov-Gerasimenko achieves 20 Kbps. New Horizons at the distance of Pluto achieves only 2 Kbps owing to the extreme range problem. MarCO achieved 16 Kbps at the distance of Mars—a rate that is quite good for a CubeSat in deep space in comparison with these much larger-class missions.

Data rates at the same range could be boosted by a factor of 8 or 9 by switching to Ka-band and by a further factor 6 by using a 1×1 m antenna. The trade-off is that the narrower transmit antenna beamwidths (associated with higher frequencies) means that pointing toward Earth can become challenging. Attitude determination and control systems (ADCS) currently available for CubeSats, however, can achieve the platform stability and pointing accuracy needed to support these high-frequency communication systems. Additionally, aggressive use of onboard data compression can achieve high science return, as proven by NASA during the Galileo mission. New Horizons made use of time, taking more than a year after the flyby to send back the highest resolution images from Pluto and Charon. Both mission architectures could well be applicable to CubeSat deep space missions.

On the horizon, optical communications will ultimately be available for the CubeSat platform. A downlink data rate of 600 Mbps was achieved by a CubeSat in LEO,

demonstrating the Optical Communications and Sensor Demonstration (OCS-D) flown by the Aerospace Corporation. A data rate of 200 Gbps may well be achieved by TBIRD, an upcoming mission by the Massachusetts Institute of Technology (MIT) and Lincoln Laboratories. The Lunar Reconnaissance Orbiter (LRO) achieved 100 Mbps using an experimental optical communication system at lunar distances.

Uplink for commanding and ranging typically requires lower data rates and can often be achieved using omnidirectional antennas aboard the spacecraft; to allow for loss of high-gain antenna (HGA) pointing (e.g., in safe mode). Several of the interplanetary CubeSats in development including Lunar IceCube, Lunar Flashlight, and LunaH-Map, utilize the same Iris X-band communication system for uplink and downlink, operating uplink at a much lower data rate. Regardless of the strategy used, a good practical guideline is to have enough bandwidth to allow for spacecraft flight software upgrades in a reasonable timeframe (e.g., one or a few DSN 8-hour passes).

5.4 Deep space navigation and tracking of interplanetary CubeSats

5.4.1 Deep space navigation and tracking of interplanetary CubeSat ground support

CubeSats in LEO can be tracked and telemetry downloaded using a ground station comprised entirely of commercially available hardware and operated by anyone with a few hours' training. Add a few thousand-dollars worth more equipment and a licensed amateur radio operator, and commands may be sent to your CubeSat. For most LEO orbits, a ground station may be sited to enable communication no less frequently than every few days. For polar orbits, communications can be enabled nearly every orbit, 90 min apart. Alternatively, equipment and services can be purchased to communicate nearly anytime through orbital relay. Rudimentary navigation services can be set up utilizing the Joint Space Operations Command website. The situation is entirely different for spacecraft going to the Moon and beyond.

While UHF frequencies, in theory, could be used at lunar distances, getting a signal back to Earth, and commands up to the spacecraft, generally requires higher frequencies to limit space losses, for example, S-, X-, and Ka-bands. Obtaining navigation information typically requires measuring Doppler shifts on already weak signals and comparing measured values with predicted values for an estimated orbit, revising the estimate, and driving down differences in an automated, iterative process that proceeds as far as data link noise allows. These functions are readily performed by equipment and people comprising NASA's Deep Space Network (DSN) and smaller networks deployed by ESA, JAXA, RSA, and ISRO. Millions of dollars worth of highly specialized equipment, including sensitive receivers operating at cryogenic temperatures, is required to receive, process, and transmit the signals required to communicate with a world fleet of a few dozen spacecraft, ranging from Voyager 1 and 2 now just barely into interstellar space beyond the Heliopause, to Parker Solar Probe's close approaches to the Sun. To enable interoperability, these networks all adhere to particular Coordinating Committee for Space Data Standards (CCSDS) protocols.

Because of the high cost, these few ground stations are heavily scheduled servicing existing interplanetary missions. While the DSN is making plans to support the 13 CubeSats to be launched with Artemis 1, allocations of time for navigational tracking of each spacecraft, accepting downlinked data and telemetry and commanding, will be minimal and subject to the needs of higher-priority missions, including Artemis 1 itself. It is notable that the DSN is opening itself to the creation of DSN affiliate stations to enable the availability of more apertures to service the anticipated increase in numbers of interplanetary CubeSat and SmallSat missions [29,30].

The first of these, DSS-17, utilizes a 21-m X-band compatible dish at Morehead State University. Other sites may be added around the world to provide greater aperture availability. For trajectories taking a CubeSat beyond a small fraction of 1 AU, data rates no more than a few kbps are viable for the power levels available on a 6U CubeSat. Thus for instruments that produce high data volumes, large onboard storage and a mission plan enabling later playback in small chunks are required. Strategies for mission operations related to interplanetary CubeSats have been developed and proven out on the MarCO mission, and the first steps toward providing tracking, ranging, telemetry, and command services for the impending Armada of deep space CubeSats have been taken [31,32] (Fig. 3).

Given the situation imposed by long distances, interplanetary CubeSats need to be ultimately imbued with significant autonomy software and hardware to (a) process any large data volumes onboard, in some cases elevating the information content per kilobit of data one hundred-fold or one thousand-fold, or more; (b) enable onboard algorithms to detect and respond to abnormal situations and minimize the need to extended “safe modes” that require extensive ground interaction to exit; (c) make decisions as to the highest priority information to return from data stored onboard (see onboard storage and data processing); and (d) (a future possibility) derive much or all of needed navigation information from onboard sources, with only occasional



Fig. 3 The Morehead State University 21-m ground station has been upgraded with support from NASA’s Advanced Exploration Systems to become the first non-NASA Affiliated node on the DSN. Referred to as DSS-17, the station will support NASA interplanetary CubeSat missions.

updates or checks using Earthbound assets. Automation of interplanetary CubeSats and their associated ground support stations is inevitable.

5.5 Navigating CubeSats in deep space

Deep space navigation enables missions to precisely target distant bodies in the Solar System and to create trajectory and orbit models to explore these targets. Navigation requires the use of trajectory and orbit models, near real-time operation and control, and precise ranging and Doppler techniques. Navigating a spacecraft to deep space locations requires a team of scientists and engineers using sophisticated spacecraft transponders, extremely large ground-based antennas, digital signal processing equipment, and precise timing equipment. These navigation processes and hardware systems needed for interplanetary CubeSats are largely the same as they are for large-scale interplanetary missions.

Navigation and flight dynamics are elements of interplanetary CubeSat missions that do not scale; they require essentially the same resources as other interplanetary missions. Trajectory and orbit modeling tools have been developed by NASA and other organizations for numeric modeling of complicated trajectories for a variety of propulsion systems, including low-thrust solar electric propulsion systems. Low-thrust trajectory design, in particular, has been optimized through a variety of analytic tools that are now available. Precision tracking requires the use of a variety of techniques including Doppler, ranging and Delta-DOR (delta-differential one-way ranging), and PN ranging (pseudonoise ranging). These techniques require a highly capable ranging transponder on the spacecraft and precision timing on the ground (typically a Hydrogen MASER). JPL successfully demonstrated the Iris transponder, a CubeSat X-band transponder system on the MarCO mission. The Iris, combined with DSN assets (including DSS-17), proved that CubeSat technology has evolved to the point of supporting deep space missions.

Future CubeSat missions will build on these successful developments to meet higher performance requirements and allow CubeSat missions to reach deeper into outer space. Future CubeSat missions, like other deep space missions, will rely on an increasing level of autonomy for navigation and will ultimately use optical time transfer (when optical communications are available). Chip-scale atomic clocks, currently under development for space applications, will improve onboard spacecraft timing and therefore facilitate increased precision in tracking and ranging. Ultimately, Pulsars may be used for deep space navigation. What is clear is that technology exists now to support interplanetary CubeSat missions and technology is on the horizon that will support missions that have not been dreamed of today.

6 Mission implementation

6.1 Success on (and lessons from) the first try: MarCO A and B

Early in 2014 the Mars Cube One (MarCO) concept was developed to independently send two small spacecraft to Mars after launching with and separating from the InSight launch vehicle. MarCO's mission, as initially conceived, was to perform

atmospheric observations through radio occultation experiments. NASA's CubeSat Challenge utilized lessons from INSPIRE to develop appropriate competitions that pushed the envelope for deep space CubeSat capability. The MarCO mission, from initial concept through interplanetary flight operations, was a technological demonstration to advance the technology necessary to bring NanoSpacecraft into deep space. While MarCO itself took advantage of the significant developments of INSPIRE (a 3U CubeSat whose intent was to demonstrate the utility of a small spacecraft in deep space, but never flew), it pioneered operational usage and laid the groundwork for many NanoSpacecraft now in development. In the end, MarCO-A and MarCO-B provided a critical real-time communication link to Earth for InSight during its entry, descent, and landing (EDL) on November 26, 2018 when InSight was out of line of sight from the Earth [33].

MarCO A and B, nicknamed Wall-E and Eva by the MarCO team, were 6U CubeSats (Fig. 4). The spacecraft were developed from commercially available components obtained from CubeSat system suppliers. The C&DH and electronics systems were provided by AstroDev of Ann Arbor, Michigan, and Blue Canyon Technologies of Boulder, Colorado, for the attitude control system (ADCS: the XACT). MMA Design LLC, of Boulder, provided solar arrays and Tyvak NanoSatellite Systems Inc., a Terran Orbital Company in San Luis Obispo, California, developed the CubeSat dispenser system. VACCO Industries of South El Monte, California, provided the cold gas micropropulsion system. The propulsion system contains eight thrusters—four canted for attitude control and four for trajectory correction maneuvers (TCMs). The propellant is R-236FA, a cold gas propellant often used in fire extinguishers [34]. The MarCO propulsion system produced 755 Ns of total impulse,

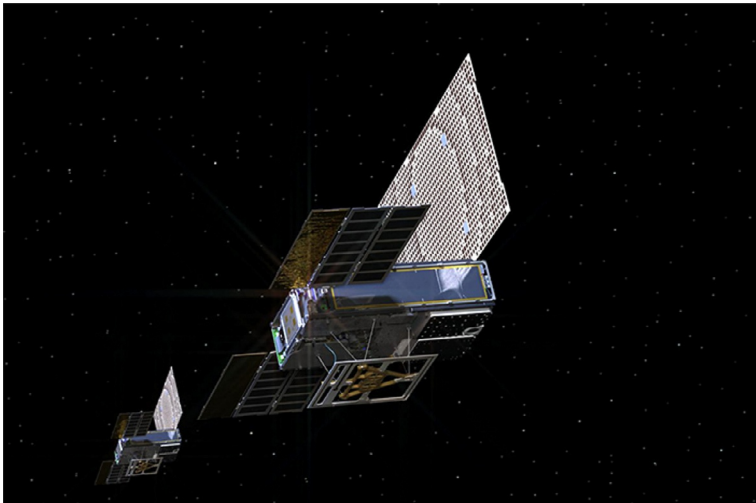


Fig. 4 Artist's conception of MarCO A and B in flight. Image courtesy NASA/JPL-Caltech.

providing in excess of 40 m/s of TCM ΔV capability. All systems were proven to be effective during deep space operations.

During the mission the MarCO spacecraft performed as intended, receiving the UHF telemetry from InSight during its EDL phase and relaying this telemetry at X-band frequencies to NASA's Deep Space Network. In addition to the InSight telemetry, the MarCO spacecraft returned an image of Mars as the spacecraft was passing 6000 km from the planet and images of the Earth–Moon system on the outbound trajectory. In all, the MarCO mission was extraordinarily successful.

MarCO's key lessons stem from the deliberate development of systems with excess capability that could be used in flight operations and the use of a small, multidisciplinary, colocated team. A streamlined governance model served to oversee and review the risks undertaken by the mission, mitigated through extensive end-to-end testing and multiple high-fidelity testbeds. The engineering model spacecraft, serving as a testbed, was critical to mission success and ultimately was used in all aspects of mission operations—checkout, nominal cruise, failure recovery, and EDL planning.

Lessons learned from operating the MarCO mission are being published in a variety of publications—two of the most significant lessons are reported here. These include compensating for a drifting onboard clock and a leaky propulsion system valve that created torques leading to momentum buildup. The MarCO team was able to find innovative solutions to these (and other) problems resulting in a highly successful mission.

Early in the mission the team discovered that the onboard clock was drifting at a significant rate, such that passes were misscheduled by seconds to minutes. Rather than directly adjust the onboard clocks, a calibration function was used to more accurately schedule spacecraft contacts. At each contact the time of carrier acquisition was recorded, along with a telemetry point for the SCLK of radio power on. Together, over multiple passes, these allowed for a simple calibration function to be created to set an onboard wake-up table and provide the DSN with appropriately scheduled contacts.

During the outbound trajectory an excessive momentum buildup was discovered and was found to be increasing over time. The reaction wheels were in danger of becoming saturated, placing the mission at risk. The team suspected that a single external valve was leaking, with the addition of the known tank-to-plenum leak; this provided a direct path for propellant to leave the spacecraft. As the leaky thruster was on a corner, it provided a net torque on the spacecraft in multiple axes, resulting in momentum buildup. The overall rate of momentum buildup was increasing, resulting in rapid and aggressive desaturations (as liquid was being expelled from the full plenum), which were unable to be compensated by the wheels. New sequences were derived to attempt to at least control the impulsive thrusting with the reaction wheels. The team attempted a fix that involved implementing a blow-down sequence every 20 min that was autonomously executed to allow the plenum to empty out of specified thrusters without further opening the tank-to-plenum valve. This would allow for controllable plenum emptying, countering the leaky thruster and keeping internal pressure low (reducing the effect of the leak). The solution worked, reserving adequate propellant, and the burns were executed effectively. The burns were observed in real time with ranging and Doppler due to the healthy power state of the spacecraft, and each segment received execution approvals following a rapid evaluation of telemetry.

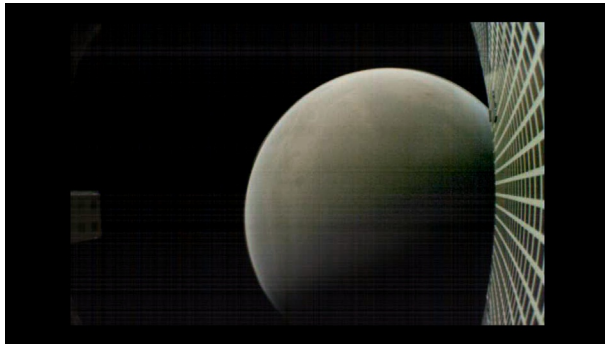


Fig. 5 By NASA/JPL-Caltech—PIA22833: Farewell to Mars on the NASA Photojournal.

Together, MarCO-A performed 9.2m/s of ΔV during TCM-1, and MarCO-B completed 6m/s, bringing both spacecraft to a trajectory with an anticipated close Mars flyby [35] (Fig. 5).

On February 5, 2019, NASA reported that both the CubeSats had gone silent and are unlikely to be heard from again. MarCO, the smallest spacecraft to ever complete an interplanetary mission, has enabled a new class of planetary exploration. At a cost forty times less than most NASA Discovery missions and a schedule of 15 months from concept to first flight article, MarCO demonstrated the feasibility of a constraint-driven interplanetary small spacecraft. The technologies developed for MarCO, including the critical Iris radio, modified commercial hardware, and the flight software, are all available for use from the mission partners and commercial entities. Continued investment from NASA, ESA, JAXA, and others implies a bright future for Solar System exploration with small spacecraft; however, as with MarCO, significant challenges, setbacks, and failures will occur.

MarCO was successful because of a narrow focus, an acceptance of risk, and significant support of its governance council. It built upon technology developed over many missions and was motivated directly from the success of the worldwide CubeSat community. While the mission was never required for InSight success, ultimately over seven million worldwide observers watched the near real-time descent of the lander through the “seven minutes of terror,” with all telemetry relayed through the two MarCO spacecraft. Even InSight’s first image of Mars was sent via MarCO; subsequent media metrics showed more than five billion “impressions” from articles, interviews, social media postings, and replays in the following days. The MarCO mission was a hard-fought success—its choices and subsequent lessons will undoubtedly inform those missions still to come.

6.2 Cislunar CubeSats

The 13 secondary payloads to be deployed on Artemis 1, following the success of MarCO, will usher in a new era of Solar System exploration with small satellite platforms. Three of the Artemis 1 lunar orbiter missions, Lunar IceCube (Morehead State

University and Partners), Lunar Flashlight (JPL), and LunaH-Map (Arizona State University and Partners) will contribute to our understanding of the lunar global water cycle and its impact on in situ surface modification processes (degree of interaction with solar wind protons resulting in physical or chemical changes on grain surfaces and the release or generation of volatiles) and on the potential generation or storage of water as a resource [36]. Table 2 describes the impact of each mission on our understanding of sources and sinks for water.

Collectively the Artemis 1 lunar orbiters will provide complimentary observations to be used in understanding volatile dynamics, addressing NASA’s Strategic Knowledge Gaps related to lunar volatile distribution (abundance, location, and transportation physics of water ice). A brief description of each mission follows.

6.2.1 Lunar IceCube

Lunar IceCube (L-IC) is a 6U CubeSat designed to prospect for water in solid, liquid, and vapor forms and other volatiles from a low-perigee, highly inclined lunar orbit. The mission was selected by NASA to fly on Exploration Mission-1 (renamed Artemis 1) in 2021. Sponsored by NASA’s Advanced Exploration Systems (AES) Program, the mission is a partnership between Morehead State University, NASA Goddard Spaceflight Center, JPL, the NASA Independent Verification and Validation Center (IV&V), and Busek Space Propulsion Company. Lunar IceCube will be deployed during translunar trajectory by the Space Launch System (SLS) and use an innovative RF ion engine to achieve lunar capture and the science orbit (inertially locked, highly elliptical, 100-km periapsis) to investigate the distribution of water as a function of time of day, latitude, and regolith composition in the context of lunar mineralogy. Lunar IceCube will include the Broadband InfraRed Compact High-Resolution Exploration Spectrometer (BIRCHES), developed by GSFC—a compact version of the successful New Horizons instrument designed with the high spectral resolution (5nm) and wavelength range (1–4μm) needed to distinguish forms of

Table 2 Comparison of volatile mission impacts on understanding the water cycle.

Water source or sink	Solar wind (time of day, latitude, shadowing effects)	Meteorite impact (transients)	Pyroclastics, other bound-water minerals (globally distributed volcanic deposits)	Polar cold traps (shadowed regions, buried ice)	Thermal migration (time of day, latitude, poleward facing slopes)
L-IC	Time of day	Maybe	Yes	No	Yes
LM	Shadowing	Maybe	No	Yes	No
LF	Shadowing	Maybe	No	Yes	No

water, including ice [37]. The mission will complement the scientific work of other missions by focusing on the abundance, location, and transportation physics of water ice on the lunar surface at a variety of latitudes. Lunar IceCube, while primarily a science mission, will demonstrate technologies that will enable future interplanetary exploration with small satellite platforms including radiation-hardened systems and subsystems, the Iris precise ranging transponder/transceiver, a capable attitude determination and control system, a high-power solar array, and an innovative electric propulsion system (EP). The EP (Busek BIT-3 Iodine engine) generates 1.2 km/s of ΔV [BIT-3] and, combined with an innovative low-energy manifold trajectory on the interplanetary superhighway, allows the spacecraft to reach lunar orbit from Earth escape with minimal energy.

The primary science objectives of the Lunar IceCube mission are to undertake spectral determination of the composition and distribution of volatiles in the lunar regolith as a function of time, latitude, and regolith age and to provide a geological context for those measurements through spectral determination of mineral components. Lunar IceCube could determine the relationship between adsorbed and bound water, hydroxyl, and ice throughout the diurnal cycle. The mission also has the potential to lend insight into understanding the role of external sources, internal sources, and solar wind proton and micrometeorite bombardment in formation, trapping, releasing of water, and exosphere formation.

Although previous missions (e.g., Clementine, Chandrayaan-1, and LRO/LCROSS) discovered various signatures of OH/H₂O on the lunar surface, they were not optimized for volatile characterization. BIRCHES is designed with a high spectral resolution (5 nm) and wavelength range (1–4 μm) needed to fully characterize water and other volatiles and to distinguish forms of water, including ice. Because the emphasis was on maximizing coverage during the nominal mission, LRO was not designed to provide repeated systematic (by time of day) measurements of representative features at higher and lower latitudes. IceCube is designed to provide these systematic measurements [38]. Working together as the first “ad hoc” constellation of interplanetary CubeSats exploring the Moon, Lunar IceCube, Lunar Flashlight, and LunaH-Map have the potential to lend significant insight into the location, depth, cyclic nature, and transport mechanisms of water on the Moon (Fig. 6).

6.2.2 Lunar Flashlight

Lunar Flashlight (LF) is a 6U CubeSat mission developed by a team from the Jet Propulsion Laboratory and the Marshall Space Flight Center that was selected by NASA’s Advanced Exploration Systems (AES) program in 2013. Planned to launch on the Space Launch System’s Artemis 1 flight in 2021, this innovative, low-cost CubeSat mission will map the lunar South Pole for volatiles and demonstrate several technological firsts, including being among the first CubeSats to reach the Moon, the first planetary CubeSat mission to use green propulsion, and the first mission to use lasers to look for water ice. Locating ice deposits in the Moon’s permanently shadowed craters addresses one of NASA’s Strategic Knowledge Gaps (SKGs) to detect

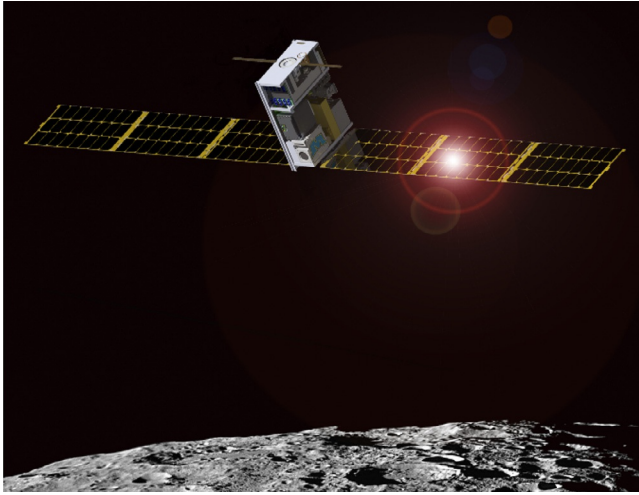


Fig. 6 The Lunar IceCube mission is designed to prospect for water ice and other lunar volatiles from lunar orbit. Morehead State University is leading the mission in partnership with NASA Goddard Spaceflight Center, JPL, Busek, and the NASA Independent Verification and Validation Center (IV&V).
Image credit: Morehead State University.

composition, quantity, distribution, form of water/H species, and other volatiles associated with lunar cold traps [39].

The scientific and economic importance of lunar volatiles extends far beyond the question “is there water on the Moon?” Volatile materials including water come from sources central to NASA’s strategic plans, including comets, asteroids, interplanetary dust particles, interstellar molecular clouds, solar wind, and lunar volcanic and radiogenic gases. The volatile inventory, distribution, and state (bound or free, evenly distributed or blocky, on the surface or at depth, etc.) are crucial for understanding how these molecules interact with the lunar surface and for utilization potential [40].

The Lunar Flashlight mission spacecraft maneuvers to its lunar polar orbit and uses its near-infrared lasers to shine light into the shaded polar regions, while the onboard reflectometer measures surface reflection and infers composition from the reflectance ratios between the different laser bands. The Lunar Flashlight 6U spacecraft has heritage elements from predecessor systems including JPL’s Interplanetary NanoSpacecraft Pathfinder in a Relevant Environment (INSPIRE), Mars CubeSat One (MarCO), and JPL’s experience with spectrometers, including the Moon Mineralogy Mapper (M3) [41]. The mission will demonstrate a path where 6U CubeSats could, at a much lower cost than previously thought possible, explore, locate, and estimate size and composition of ice deposits on the Moon. Polar volatile data collected by Lunar Flashlight data could be key to selecting future surface exploration targets, for more expensive lander- and rover-borne measurements, based on its laser reflection measurements of surface ice fraction and areal extent (Fig. 7).

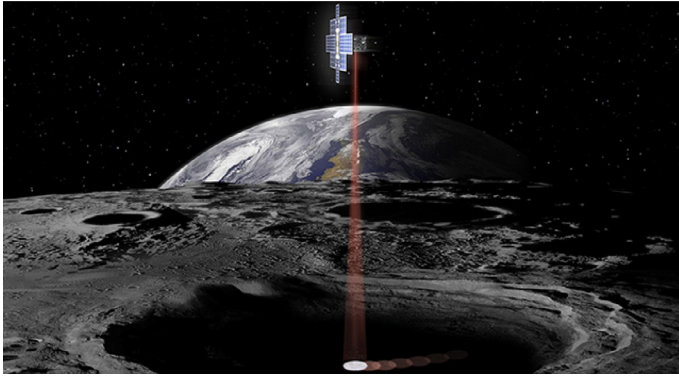


Fig. 7 Artist's concept showing the Lunar Flashlight spacecraft, a six-unit CubeSat designed to search for ice on the Moon's surface. The spacecraft will use its near-infrared lasers to shine light into shaded polar regions on the Moon, while an onboard reflectometer will measure surface reflection and infer composition.

Image credit: NASA/JPL-Caltech.

6.2.3 *LunaH-Map*

The Lunar Polar Hydrogen Mapper (LunaH-Map, or LM) is a 6U+ CubeSat funded through NASA Science Mission Directorate's Small, Innovative Missions for Planetary Exploration (SIMPLEX) program. LunaH-Map will make maps of hydrogen enrichments near the Moon's South Pole at spatial scales smaller than the extent of the neutron suppressed regions detected by the NASA Lunar Prospector (LP) and Lunar Reconnaissance Orbiter (LRO) missions [42]. Like LP and LRO, LunaH-Map will use a neutron spectrometer to create a map of hydrogen abundance; however, LunaH-Map will fly at much lower altitudes over the lunar South Pole to create maps of hydrogen enrichments at finer spatial resolutions than achieved by previous missions. Arizona State University began the development of LunaH-Map after being awarded a contract by NASA in early 2015. LunaH-Map's primary objective is to map the abundance of hydrogen down to 1 m beneath the surface of the lunar south polar region. This information may then be used to improve scientific understanding of how water is initially distributed and spread throughout the Solar System and where water might be found by future manned missions for life support and fuel production. It will be inserted into a polar orbit around the Moon, with its periapsis located near the lunar South Pole, initially passing above Shackleton crater [43].

Planetary nuclear spectroscopy relies on detection of leakage neutrons generated by galactic cosmic rays interacting within the top meter of the body's surface. Fast neutrons produced by the GCR interaction undergo further interactions with the regolith. With a mass roughly equivalent to one neutron, hydrogen is particularly efficient at reducing the energy of neutrons. As such, regions with increased hydrogen abundance will have a depressed flux of epithermal ($E > 0.3$ eV) neutrons, whereas the flux of thermal ($E < 0.3$ eV) neutrons will be enhanced. The mission's primary payload, the

Miniature Neutron Spectrometer (Mini-NS), has been designed to use the scintillator material $\text{Cs}_2\text{YLiCl}_6\text{:Ce}$ and a thin shield of gadolinium (which absorbs thermal neutrons) to measure count rates of epithermal neutrons [44].

6.3 Asteroids, Mars, and the outer Solar System

Concepts for interplanetary CubeSat missions have stretched beyond the Moon to include the inner Solar System, asteroids, and ultimately the outer Solar System. The first of these missions likely to be implemented since MarCO is the Near-Earth Asteroid Scout (NEA Scout). NEA Scout, one of the 13 missions selected for Artemis 1, is a 6U CubeSat developed by a partnership between NASA's Marshall Space Flight Center and the Jet Propulsion Laboratory. NEA Scout is a flyby encounter of an asteroid representative of Near-Earth Asteroids (NEAs) that are potential human destinations. Target NEAs are currently being evaluated and are updated based on the dynamic launch date of Artemis 1, on new discoveries and expected performance. The current planned target is 1991VG, a small NEA in a heliocentric orbit that brings it near the Earth (0.0568 AU in 2017), making it a prime target for CubeSat exploration [45].

NEA Scout's science mission is to determine the characteristics of 1991VG including its global shape, spin rate, pole position, regional morphology, regolith properties, and spectral class using an optical imager to capture a series of low- (50 cm/pixel) and high-resolution (10 cm/pixel) images. The mission's concept of operations is unique. SLS will carry the CubeSat to Earth escape and placed on lunar trajectory. Once NEA Scout reaches the lunar vicinity, it will utilize a cold gas propulsion system to provide the initial propulsive maneuvers. NEA Scout will then utilize an 84 m² solar sail, leveraging the CubeSat's continual solar exposure for efficient transit to the target asteroid during an approximate 2-year cruise [46]. NEA Scout, along with the other Artemis 1 CubeSats, will prove out innovative enabling technology to support future Solar System exploration utilizing the CubeSat form factor and produce science data that will ultimately support human exploration of the Solar System (Fig. 8).

A recent study conducted jointly by JPL and the Aerospace Corporation looked at microlanders that could be carried to Mars as secondary payload(s) on spacecraft bound for Mars [47]. In the study, each lander (Fig. 9) consists of a 30-cm-diameter probe, carrying a ~1 kg scientific payload. The concept built on the success of Aerospace's small Earth Reentry Breakup Recorder (REBR) vehicle (which was a CubeSat-based technology demonstration) to design a low mass/low ballistic coefficient entry system that allows for subsonic deployment of a steerable parawing hang glider, capable of up to 10 min and more than 10 km of guided flight, which would impact the surface with a 3:1 glide ratio at ~20.5 m/s. The MarsDROP concept would enable scientists to reach areas of Mars that are inaccessible to larger landed missions [47].

In October 2014, JPL selected proposals from 10 universities to study CubeSat concepts that could enhance the Europa Clipper mission currently planned for a 2023 launch opportunity by NASA [48]. As illustrated in Fig. 10, such CubeSats would be carried to Jupiter by the host spacecraft and then released on approach to Europa to execute their assigned mission. In this mother-daughter architecture, the CubeSat

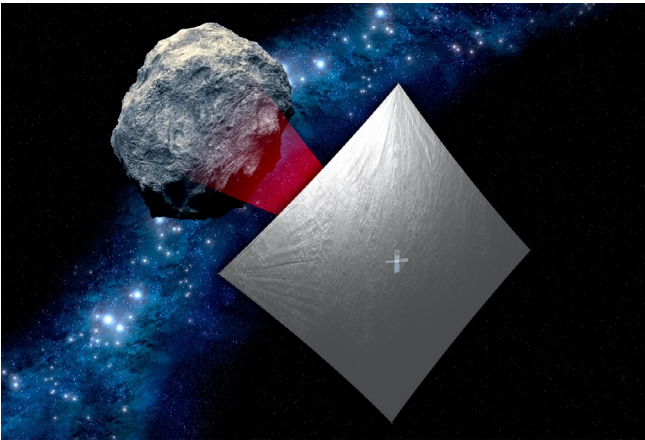


Fig. 8 Artist’s conception of NEA Scout CubeSat with its solar sail deployed as it characterizes a near-Earth asteroid. Image is courtesy of NASA.

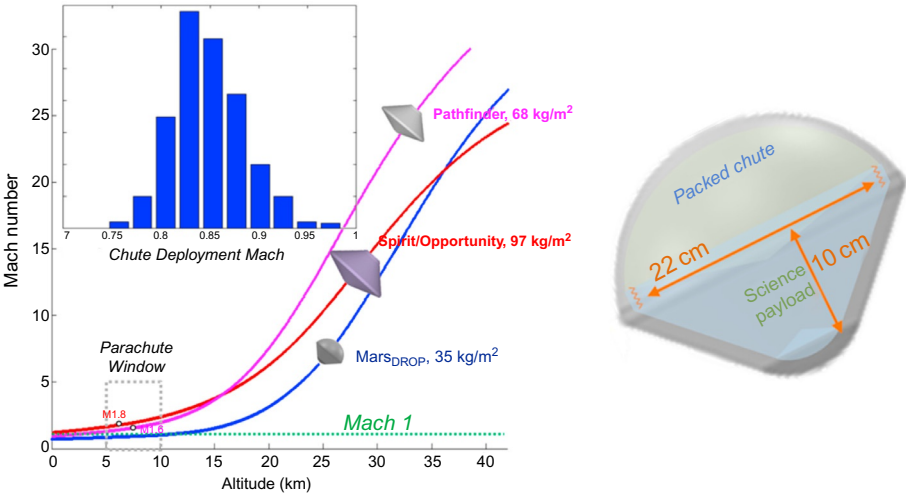


Fig. 9 MarsDROP MicroProbe concept.

elements can approach much closer to the surface of Europa than the primary spacecraft, enabling unique science observations. As an example, one CubeSat under study by the University of Michigan proposed to use multifrequency magnetic induction sounding from a compact magnetometer to characterize the subsurface ocean of Europa [49]. At the time of writing, it appears that Europa Clipper will likely not have sufficient mass margin to accommodate any of the 10 concepts, but that does not eradicate the validity and creativity of these mother-daughter concepts, which can be dusted off and revamped for future outer planet missions (Fig. 10).

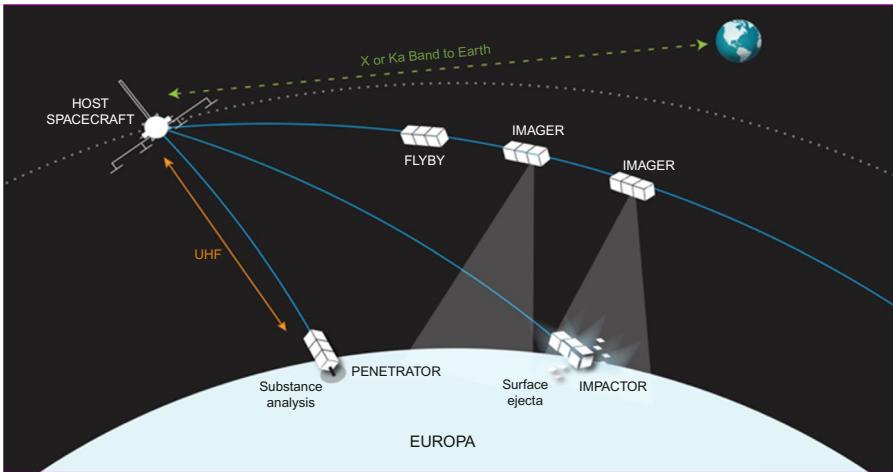


Fig. 10 Illustration of CubeSats deployed from the Europa Clipper mission.

6.4 Venus

Recent mission concept studies continue to push the boundaries of the science that is possible with CubeSats and SmallSats. Another recent study at JPL examined a free-flyer nanosat probe, launched directly from Earth toward Venus [50]. While the probe does not conform strictly to the CubeSat form factor, it is highly derivative of CubeSat systems. The probe targets very high priority science at Venus—the need to measure the relative abundances of Neon, Argon, Helium, Krypton, and other noble gases to understand how Venus’ atmosphere formed and has evolved. To be truly representative of the noble gases and their isotopic ratios, a sample of the atmosphere has to be acquired where the atmosphere is well mixed at an altitude below the homopause, which for Venus is around 120 km.

The study came up with a novel, low-cost, SmallSat architecture for a conceptual stand-alone mission that would sample the noble gases and their isotopic ratios at Venus. Sampling is achieved by a compact ion-trap mass spectrometer incorporated into a small (60 cm diameter) probe that skims through the atmosphere, targeting a closest approach altitude above the surface below the nominal 120 km (Fig. 11). Following acquisition the gas sample is analyzed over a period of ~60 min.

The Venus probe and carrier spacecraft are assumed to launch together on a Type II trajectory toward Venus, on a dedicated smaller-class rocket, with a kick stage. The probe and carrier spacecraft both make extensive use of NanoSpacecraft components. The compact mass spectrometer instrument mass is estimated at 8 kg. En route to Venus the carrier spacecraft would execute preplanned TCM maneuvers for a total ΔV of about a hundred meters per second. On approach the carrier spacecraft would spin up to rotate at ~10 rpm and then release the probe on its path to skim through the atmosphere. The carrier spacecraft would then execute a small maneuver (with a few meters per second of ΔV) to fly past Venus above the atmosphere. Each sample

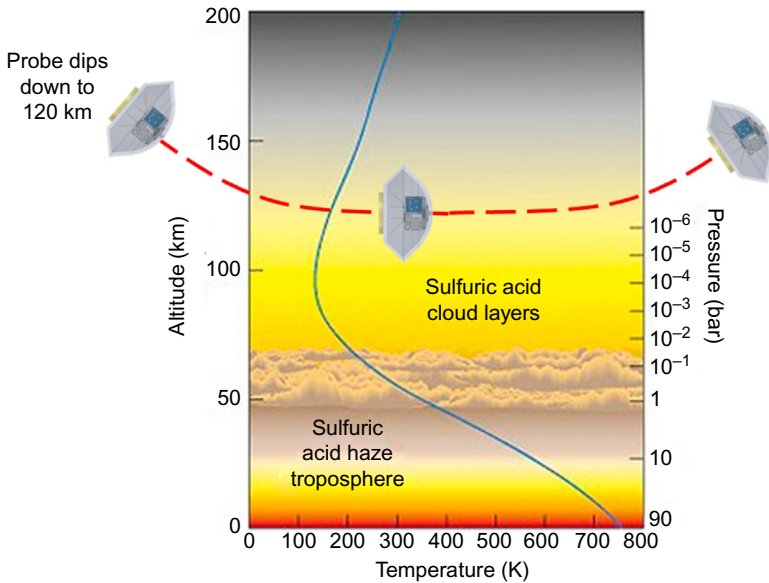


Fig. 11 Venus atmospheric sampling probe would skim through the atmosphere to sample the noble gases below the homopause at ~ 120 km altitude above the surface. Image courtesy of NASA/JPL/Caltech.

acquired is then analyzed by a miniaturized quadrupole ion trap mass spectrometer (QITMS) developed at JPL (see Fig. 12). A calibrant tank provides a reference for calibration. Data are transmitted to Earth during the long apoapsis segment of the orbit. The current estimate for the mass of the dry probe is 70 kg, including margins. Following the probe skim through and data analysis, results are returned to Earth via a UHF/X-Band relay, similar to that developed for MarCO.

7 Planned NASA interplanetary CubeSat missions

Today, Solar System exploration missions are the exclusive domain of space agencies and their scientists and engineers who can muster multi-hundred-million dollar budgets. While their accomplishments are broad, the high cost limits our pace of important discoveries. Interplanetary CubeSats, however, offer an opportunity to conduct focused science investigations around the inner Solar System at a cost 10 times lower than missions mounted today. Adopting this exploration philosophy, NASA has dozens of interplanetary CubeSat missions in various stages of planning and development. Among these are numerous missions planned by JPL, NASA Centers including the Goddard Spaceflight Center, the Marshall Space Center, and a collection of missions funded as studies under the NASA Planetary Science Deep Space SmallSat Studies (PSDS3) program.

Venus NanoSat probe concept

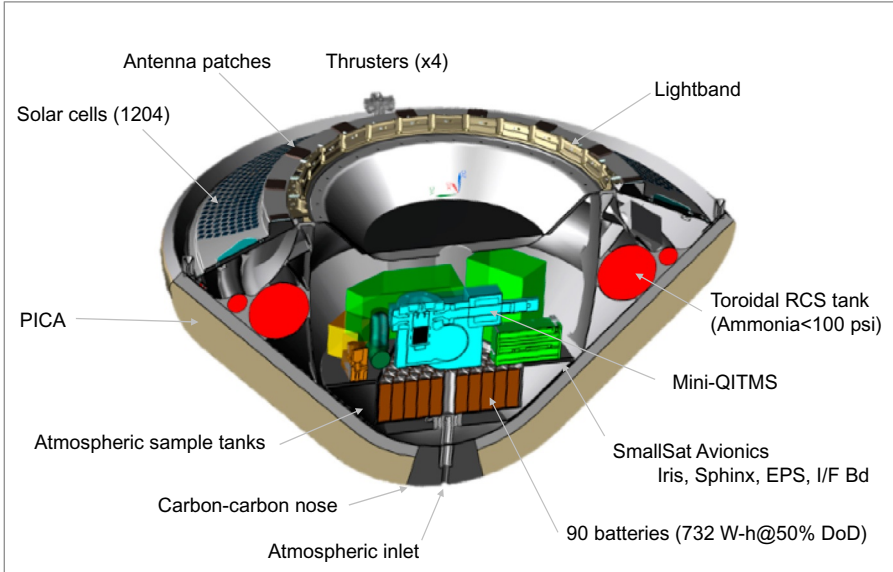


Fig. 12 Cutout depicting the design of the Cupid's Arrow Venus atmospheric "skimmer". Image courtesy of NASA/JPL/Caltech.

Ten studies were selected under the PSDS3 program that utilize SmallSat platforms to explore Venus, Earth's Moon, asteroids, Mars, and the outer planets. For the purpose of the study, SmallSats were defined as having a mass less than 180 kg [51]. Several of the PSDS3 missions are CubeSats; these are briefly described in the succeeding text, categorized by the Solar System body targeted for exploration.

7.1 Venus

7.1.1 CubeSat UV Experiment (CUVE)

CUVE is a study undertaken at the University of Maryland, College Park based on a 12-unit CubeSat orbiter that is intended to measure ultraviolet absorption and night-glow emissions to understand Venus' atmospheric dynamics.

7.1.2 Cupid's Arrow

Cupid's Arrow is a 70-kg probe proposed by JPL to measure noble gases in the atmosphere of Venus and their isotopes to investigate the comparative geological evolution of Venus and the Earth (described previously).

7.2 Earth's Moon

7.2.1 CubeSat X-ray Telescope (CubeX)

CubeSat X-ray telescope (CubeX) is a study undertaken by the Smithsonian Astrophysical Observatory, Cambridge, Massachusetts. The primary mission of CubeX, a 12-unit CubeSat, is to map the elemental composition of airless bodies such as the moon to understand their formation and evolutionary history.

7.2.2 Bisat Observations of the Lunar Atmosphere above Swirls (BOLAS)

BOLAS is a collaboration between the NASA Goddard Spaceflight Center, Morehead State University, and Tethers Unlimited to perform a study to deploy two 12U CubeSats in lunar orbit tethered by a thin tens of kilometers-long tether allowing them to fly in a gravity gradient formation. The objective is to investigate the hydrogen cycle by investigating the mechanisms and dynamics of lunar hydrogen implantation and their dependence on composition, regolith properties, local topography, plasma conditions, time of day, and crustal magnetic fields [52].

7.3 Asteroids

7.3.1 Asteroid Probe Experiment (APEX)

APEX is a study undertaken by the Johns Hopkins University Applied Physics Laboratory, Laurel, Maryland to develop a SmallSat with a deployable seismometer to rendezvous with the asteroid Apophis and directly explore its interior structure, surface properties, and rotational state.

7.3.2 CubeSat Asteroid Encounters for Science and Reconnaissance (CAESAR)

Lockheed Martin Space Systems Company, Littleton, Colorado led the CubeSat Asteroid Encounters for Science and Reconnaissance (CAESAR) study; CAESAR is a constellation of 6-unit CubeSats designed to evaluate the bulk properties of asteroids to assess their physical structure and to provide constraints on their formation and evolution.

7.4 Mars

7.4.1 Chariot to the Moons of Mars

Purdue University, West Lafayette, Indiana led the Chariot to the Moons of Mars study program. Chariot is a 12-unit CubeSat with a deployable drag skirt designed to produce high-resolution imagery and surface material composition of Phobos and Deimos to lend insight into the formation of these Moons of Mars.

7.4.2 *Aeolus*

NASA Ames Research Center, Moffett Field, California led the Aeolus study. The Aeolus mission concept is based on a 24-unit CubeSat intended to directly measure vertically resolved global winds to help determine the planetary-wide energy balance of Mars and to lend insight into Martian climate variability.

7.5 *Icy Moons and outer planets*

7.5.1 *Small Next-generation Atmospheric Probe (SNAP)*

The SNAP study was led by team at Hampton University, Virginia. SNAP is an atmospheric entry probe intended to measure vertical cloud structure, stratification, and winds of the atmosphere of Uranus. The science goal is to lend insight into the chemical and physical processes that shape the atmosphere of Uranus.

7.5.2 *JUPiter Magnetospheric boundary Explorer (JUMPER)*

The JUMPER concept study was led by Southwest Research Institute, San Antonio, Texas. JUMPER is a SmallSat intended to explore Jupiter's magnetosphere, including characterization of the solar wind upstream of the magnetosphere to providing science context for future missions such as the Europa Clipper.

While the missions described in the preceding text do not represent a comprehensive list of NASA interplanetary CubeSat missions in study and planning, the number and breadth of these missions illustrate the level at which NASA has embraced the interplanetary CubeSat paradigm.

8 *Planned ESA interplanetary CubeSat missions*

The European Space Agency has also enthusiastically adopted the CubeSat paradigm for some of its upcoming Interplanetary missions, as described in this section.

8.1 *CubeSats on the Hera mission to Didymos*

ESA's Hera mission to the binary near-Earth asteroid Didymos is the agency's contribution to the international Asteroid Impact and Deflection Assessment (AIDA) collaboration with NASA. The NASA Double Asteroid Redirection Test (DART) spacecraft planned to launch in late 2020/early 2021 is targeting to impact the smaller of the two bodies in the Didymos binary system (dubbed "Didymoon") at very high velocity in late 2022. Hera is under development for launch in 2024 and is planned to rendezvous with Didymos in 2026 to perform a detailed characterization of Didymoon postimpact and hence gather important data relevant to the kinetic energy impact technique for asteroid deflection in planetary defense scenarios. As part of its payload, Hera will carry two 6U CubeSats on its interplanetary cruise phase and deploy them upon arrival at Didymos. During the long cruise the CubeSats are stowed inside a CubeSat Deployer specially developed to provide power, data, and thermal interfaces.

Due to the weak gravity field of the asteroid, the Deployer has to be capable of deploying the two CubeSats at very low velocity of only a few cm/s; otherwise they will quickly drift away from the asteroid vicinity. The two CubeSats will be operated via an S-band intersatellite link (ISL) with the Hera mothercraft as a communication relay with Earth ground stations and mission control. Once deployed at the Didymos system, the CubeSats are planned to be navigated in close proximity to the smaller body Didymoon where they will perform close-up measurements with their respective miniaturized instrument suites and eventually attempt a landing. Navigation is enabled by a combination of ranging function over the ISL and visual camera-based center-of-brightness observations relative to Didymain and Didymoon [53].

The first CubeSat is called Juventas, developed by Danish company GomSpace and GMV in Romania. It will measure the gravity field and the internal structure of the two smaller Didymos asteroids. In close orbit around Didymoon, Juventas will line up with Hera to perform satellite-to-satellite radio science experiments and carry out a low-frequency radar survey of the asteroid interior, similar to performing a detailed “x-ray scan” of Didymoon to unveil its interior. The adventure will end with a landing using the dynamics of any likely bouncing to capture details of the asteroid’s surface material, followed by several days of surface operations (Fig. 13). Due to funding constraints, the second CubeSat is currently under consolidation and is planned to start development in the short term.

8.2 *Lunar CubeSats for Exploration (LUCE)*

Within the framework of the European Exploration Envelope Programme (E3P), a number of commercial partnerships have been established between ESA and industry. One of these is the Commercial Lunar Mission Support Services partnership with



Fig. 13 The Juventas CubeSat will probe the interior of the Didymos asteroid using a low-frequency radar.

Credit: ESA/Gomspace.

SSTL and Goonhilly Earth Station in the United Kingdom. The definition phase of the partnership is ongoing, and the second phase is proposed to comprise the Lunar Pathfinder mission to demonstrate provision of communication and navigation services to lunar orbiting or lander assets. Lunar Pathfinder is planned to launch in 2023 and during its operational lifetime, is planned to communicate with two 12U CubeSats in lunar orbit. After the CubeSats are deployed, they are planned to be operated by using Pathfinder as a communication data relay to the Goonhilly station interfaced to the two CubeSat mission control centers.

8.3 LUMIO

The two CubeSats were selected via an open competitive Request for Information process for Lunar Payloads. The mission concepts were previously joint winners of the ESA SysNova Challenge on “Lunar CubeSats for Exploration (LUCE),” an open competitive Announcement of Opportunity process that downselected four lunar CubeSat concepts for further study.

LUMIO is led by an Italian/Dutch/Norwegian consortium (Politecnico Di Milano, Innovative Solutions In Space b.v., TU Delft, EPFL, S[&]T Norway, Leonardo S.p.A, Uni. Arizona). The mission objective is to perform meteoroid impact flash monitoring of the lunar far side from an Earth-Moon L2 halo orbit using a high-frame rate optical camera. The CubeSat system consists of a single 12U with propulsion for L2 transfer and capabilities of autonomous onboard optical navigation of the lunar disk (Fig. 14).

8.4 VMMO

VMMO is led by a Canadian/UK consortium (MPB Communications, Surrey Space Centre, Uni. Winnipeg, and others) and is aimed at measuring the content of water ice deposits in permanently shadowed craters at South Pole, minerals (e.g., ilmenite) on



Fig. 14 Illustration of the LUMIO CubeSat at Earth-Moon L2 Lagrangian point.
Credit: Politecnico di Milano.

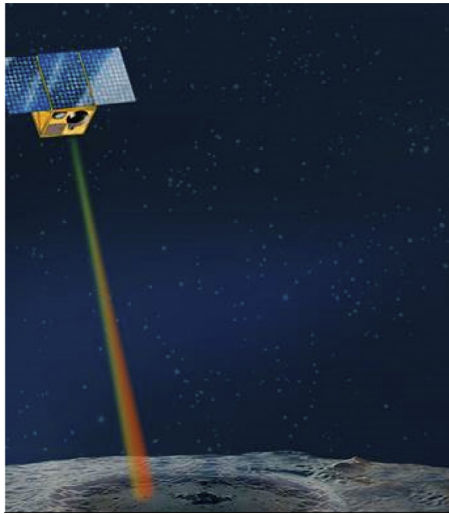


Fig. 15 Illustration of VMMO CubeSat performing low-altitude observations of lunar ice with active fiber laser spectrometer.

Credit: MPB Communications.

the day side, and the lunar radiation environment. The payload consists of an Active Fiber Laser spectrometer (1560 and 530nm with 10-m spot size) and a radiation environment sensor. The CubeSat system consists of a single 12U CubeSat with electric propulsion for transfer to very low lunar orbit for conducting the required observations at low altitude (Fig. 15).

8.5 Miniaturized Asteroid Remote Geophysical Observer (M-ARGO)

The M-ARGO mission is proposed as a highly innovative mission to demonstrate critical technologies and operations for stand-alone deep space CubeSats in the relevant environment, rendezvous with a near-Earth object (NEO), and perform physical characterization of the object with a small payload suite for in situ resource exploration purposes. A successful demonstration of M-ARGO would be the world's first CubeSat to independently rendezvous with and study an asteroid over an extended period in close proximity. ESA studies on the M-ARGO mission concept have shown that it is feasible to integrate a number of enabling miniaturized technologies (see Fig. 16) into a 12U CubeSat platform and, based on a piggyback launch to Sun-Earth L2 transfer, have sufficient performance to rendezvous with a near-Earth object while carrying a small payload. Up to 140 different NEOs were found (so far) to be accessible with the current system design. M-ARGO is planned to lower the entry-level cost of deep space exploration by over an order of magnitude. This will enable new system architectures and missions, such as fleets of nanoprobes for in situ resource exploration of NEOs and for other applications such as planetary science, planetary defense,

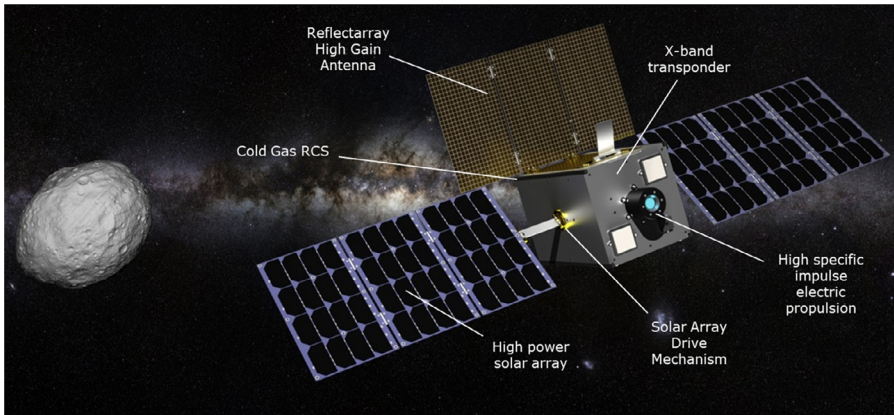


Fig. 16 M-ARGO stand-alone CubeSat mission for NEA rendezvous.

Credit: ESA.

and space weather monitoring, for example. Currently all of the critical technologies are under development. The contract for the industrial Phase A study was kicked off with GomSpace Luxembourg and Politecnico di Milano. Flight readiness is targeted for 2023, dependent upon further funding for the implementation phase [54].

9 Future opportunities

9.1 Daughter spacecraft to larger missions

To get to the outer Solar System, as discussed earlier, CubeSats are going to have to hitch a ride on larger planetary science missions, making good use of excess mass margin. One recent mission, Cassini, made countless discoveries at Saturn furthering our knowledge of the Solar System formation and finding new ocean worlds that might have the possibility of supporting life. But now that the mission has ended, questions remain unanswered, and new ones have arisen. To address this the planetary science community has proposed several follow-on missions. Dragonfly, selected for NASA's next New Frontiers mission opportunity, aims to land a quadcopter on the surface of Titan to study prebiotic chemistry and potential habitability [55]. If the mission, as selected, has ample mass margin, it could accommodate ride-along CubeSats that could accompany the primary spacecraft to Saturn and perform unique and ground-breaking science as well as demonstrate new technologies.

In a recent study [56], 10 “hitch-hiker” CubeSat and SmallSat concepts were studied that could ride along on NASA's next mission to the Saturn system. These concepts included probes similar to Cupid's Arrow but targeting the atmospheres of Venus (en route to Saturn), Saturn, and Titan, orbiting CubeSats to study the magnetosphere, stereo mapping of the South Pole of Enceladus and other phenomena at Titan, and finally a “Ring-Diver” to get the closest possible view of Saturn's famous ring system; see Fig. 17. These concepts for exploring the Saturn system would take

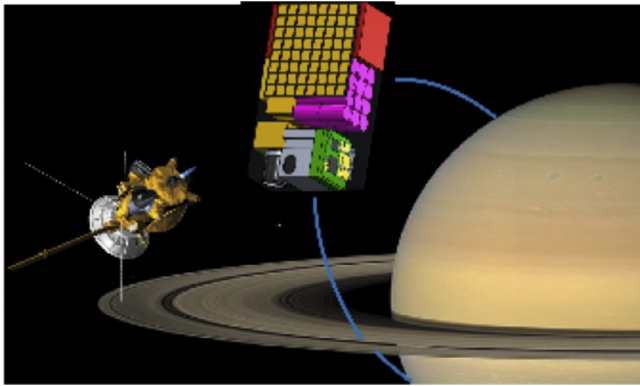


Fig. 17 The Saturn Ring-Diver CubeSat mission concept.

advantage of advances in Small/CubeSat technology, demonstrating their ability to be used outside the traditional confines of LEO. New technology and instruments would also be demonstrated for use in future missions.

JPL recently proposed seven CubeSat/Nanosat missions as technology demonstrations to augment larger missions proposed to NASA's Discovery program [57]. Each CubeSat or Nanosat was carried in a mother-daughter configuration to its destination in the Solar System, which ranged from Venus to main-belt asteroids, a Jovian comet, and Phobos [58]. Each provided a unique capability that could have augmented the science of the companion (mother) discovery mission. Some were flybys of the target body to offer a closer look than could be risked with the main spacecraft, others provided in situ measurements. None of these concepts were selected for flight, but they remain documented as evidence of the creativity of the CubeSat community, and can easily be pulled off the shelf as the foundation for future Interplanetary CubeSat missions.

One forward-looking mission concept submitted to the recent NASA Discovery call proposed to achieve all of its science using NanoSats. The NANOSWARM mission concept [59] positions a fleet of NanoSats in orbit around the Moon to address science questions on space weathering; the origins of planetary magnetism; the origins, distributions, and migration processes of surface water on airless bodies; and the physics of small-scale magnetospheres. The NANOSWARM concept uses a novel "mother-with-many-children" architecture to place its CubeSat Armada into a low, circular, polar lunar orbit. The mother ship releases some CubeSats on impact trajectories into the hearts of lunar magnetic anomalies to measure magnetic fields and proton fluxes, in real time, up until the last tens of milliseconds. A second set of CubeSats would then be released into a polar orbit with a periapsis over the South Pole to measure neutron fluxes.

In LEO, SmallSats and CubeSats have changed the exploration paradigm, offering a fast and low-cost alternative to traditional space vehicles. These small spacecraft have spawned a revolutionizing industry for performing cutting-edge science. This new mission development philosophy has the potential to significantly change the

economics of interplanetary exploration. A number of missions are in development, which utilize CubeSat class spacecraft beyond Earth orbit, while mind-blowing newer concepts will emerge inspired by the experiences of these first CubeSat Solar System explorers.

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